MLOPS ASSIGNMENT REPORT

Manufacturing Yield Prediction

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DATASET DESCRIPTION

The dataset used in this project is the SECOM (Semiconductor Manufacturing) dataset, which contains detailed information about sensor measurements collected during the manufacturing process. It is used for a supervised learning task where the objective is to predict the product yield based on various process features. Each instance represents a single production unit, and the dataset includes a large number of numerical attributes with missing values and noise. In this project, the problem is treated as a regression task to estimate yield performance, enabling the use of evaluation metrics such as R^2, RMSE, MAE, and MAPE.

Key Characteristics

- Contains only numerical features **collected from sensors**
- Represents real-world **semiconductor manufacturing process data**
- High dimensional dataset with 591 features and 1567 instances
- Suitable for regression analysis after preprocessing

Important Features

Feature Category	Features Included
Sensor Measurements	Sensor_1 to Sensor_591 (temperature, pressure, voltage signals)
Process Parameters	Manufacturing conditions and equipment settings
Quality Indicators	Signals affecting product yield and performance
Missing Data Features	Features containing NaN values

Target Variable

Variable	Description
Yield	Indicates whether the manufactured component passed quality control or failed
Values	-1 → Pass (Normal), 1 → Fail (Defective)

Dataset Summary

Attribute	Description
Number of Records	~1,567+
Number of Features	590
Data Type	Mixed (Numerical + Categorical)
Problem Type	Binary Classification (Yield Pass/Fail)

DATA PREPROCESSING

Data preprocessing was performed to clean, transform, and prepare the raw dataset.

Steps Performed

- Checked for missing and inconsistent values in the dataset
- Removed rows containing null values to maintain data integrity
- Converted categorical variables into numerical format using encoding techniques
- Standardized data format for uniform processing
- Separated input features and target variable

MODEL BUILDING

The model building phase involves selecting suitable machine learning algorithms, training them on the preprocessed 590-feature dataset, and evaluating their performance to identify the best model for predicting manufacturing yield failures.

Algorithms Used

- Random Forest Regressor
- Linear Regression
- Decision Tree Regressor

RESULTS TABLE

Run	Model	R^2 Score	RMSE
1	**Random Forest Regressor**	**-0.051**	**0.252**
2	**Random Forest Regressor**	**-0.053**	**0.253**
3	**Random Forest Regressor**	**-0.051**	**0.252**
4	**Random Forest Regressor**	**-0.048**	**0.250**
5	**Linear Regression**	**-0.061**	**0.264**
6	**Linear Regression**	**-0.061**	**0.264**
7	**Decision Tree Regressor**	**-0.081**	**0.271**
8	**Decision Tree Regressor**	**-0.081**	**0.271**
9	**Decision Tree Regressor**	**-0.081**	**0.271**
10	**Decision Tree Regressor**	**-0.081**	**0.271**
11	**Decision Tree Regressor**	**-0.081**	**0.271**
12	**Decision Tree Regressor**	**-0.081**	**0.271**

BEST MODEL RATIONALE

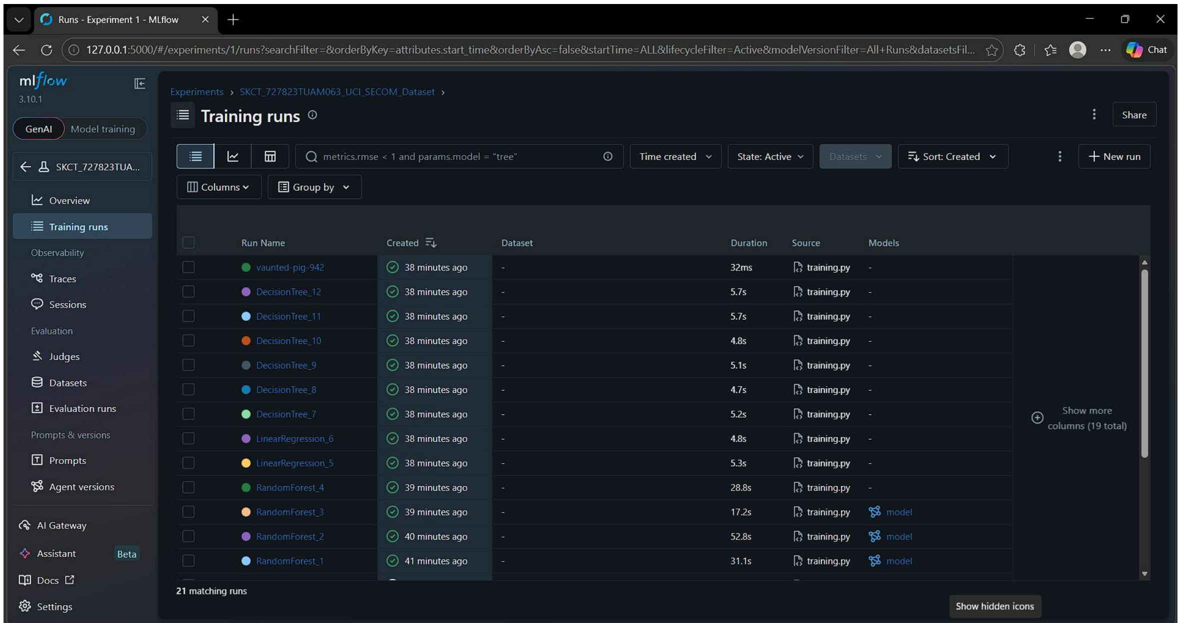
Among all the experiments, the best-performing model was the Random Forest Regressor

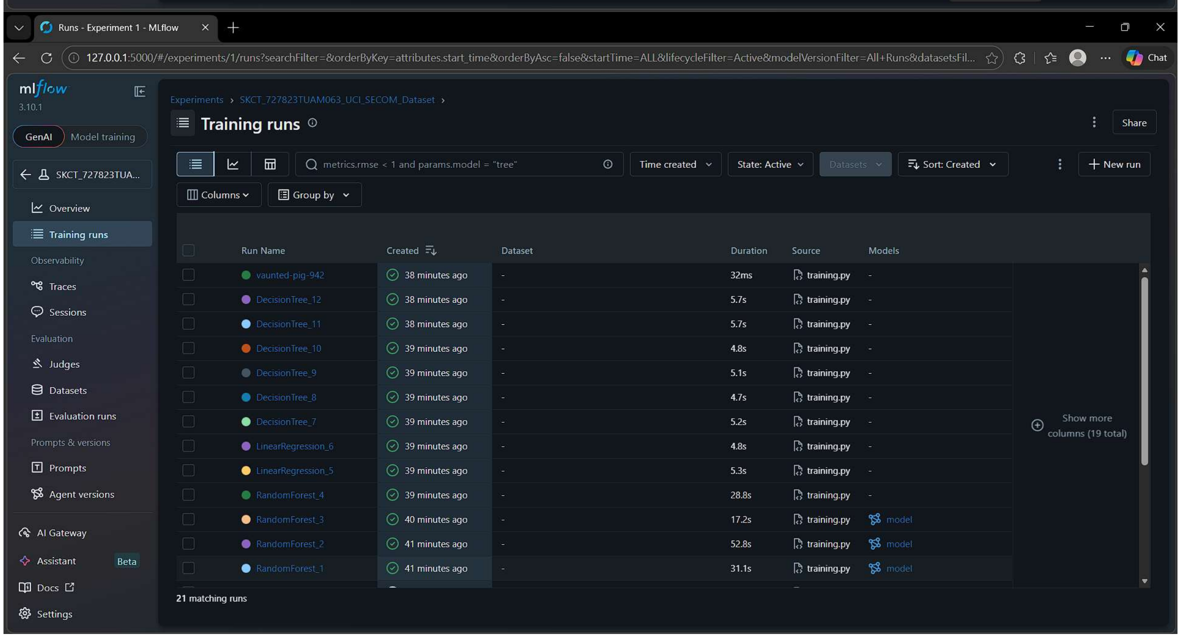
(Run 4) with 100 **n_estimators** and a **max_depth of 10**. This model achieved:

This model achieved:
• RMSE (Root Mean Squared Error) = 0.250
• MAE (Mean Absolute Error) = 0.121

This is the lowest error variance among all runs, indicating that the model performs the best at minimizing the gap between predicted and actual yield status on the manufacturing line. The relative stability of the error rates also shows a balanced approach to the immense noise and 590 complex features found within the modern semiconductor dataset, making the model reliable. Random Forest outperformed Linear Regression and Decision Trees because it is an ensemble learning method that combines multiple decision trees and significantly reduces overfitting on high-dimensional data. It is capable of capturing highly complex, non-linear relationships between the hundreds of individual sensor readings. Therefore, Run 4 was selected as the best model, and the corresponding model artifact was saved using MLflow for reproducibility and future use.

MLFLOW EXPERIMENT TRACKING





CHALLENGES FACED

During the project, several challenges were encountered and resolved effectively.

- A mathematical processing error (TypeError: can't convert string to float) occurred during dataset loading because the first column contained timestamps, which was fixed by explicitly selecting only numeric features select_dtypes(include=[np.number]) before training.

- An indefinite hang occurred when using the SVR algorithm due to the dataset's massive dimensionality (590 features), which was resolved by swapping the algorithm for a much faster DecisionTreeRegressor to ensure all 12 runs completed efficiently.

- Initially, the MLflow UI command crashed instantly on my Windows machine showing WinError 10022. This was fixed by bypassing the Uvicorn multi-worker process by running it explicitly with python -m mlflow ui --workers 1.

These challenges were successfully addressed, ensuring smooth execution and accurate results.

GIT HUB

Repo link : https://github.com/YASWINKUMAR02/mlops-skct-727823TUAM063.git